

# Testing & Quality Assurance

*Book a Doctor Healthcare Platform • Folder 5: Testing Phase*

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| <b>Project Lead:</b>    | <b>Boya Praveen</b>                                                           |
| <b>Prepared By:</b>     | Uppara Ganesh                                                                 |
| <b>Project Team:</b>    | Boya Praveen, Shaik Baba Fakruddin, Uppara Ganesh, R Sai Ram, Saginala Sushma |
| <b>Submission Date:</b> | July 2026                                                                     |

## Folder 5 Metadata & Registry

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| Registry Parameter | Value Specification                                            |
|--------------------|----------------------------------------------------------------|
| Project Stage      | Folder 5 – Testing & Validation                                |
| Primary Objective  | API Verification, UI Bug Hunting, and Performance Benchmarking |
| Status             | Validated - System Ready for Final Deployment                  |

### 1. Testing Phase Executive Summary

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Following the implementation of the MERN stack modules, Folder 5 serves as the Quality Assurance layer. This phase ensures that the "Book a Doctor" application meets all functional requirements (scheduling, browsing, secure uploads) and maintains stable performance under standard operational loads.

### 2. Quality Assurance Methodology

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| Test Module           | Verification Objectives                                                                          |
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| API Logic (Postman)   | Verify CRUD endpoints (POST /api/book, GET /api/doctors) return successful 200/201 status codes. |
| User Acceptance (UAT) | Validate seamless booking flow from patient discovery to appointment confirmation.               |
| Database Integrity    | Ensure user and appointment data are accurately reflected in MongoDB collections.                |

### 3. Bug Mitigation & Performance Benchmarks

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- **Latency:** API response times optimized to sub-500ms for routine lookups.
- **Security:** JWT implementation tested against unauthorized access requests.
- **Frontend Stability:** Component re-rendering issues resolved to ensure smooth transition between doctor profiles.

## 4. Folder 5 Conclusion

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The "Book a Doctor" application has passed all testing modules. The system is validated as accurate, secure, and performant, moving the project successfully into final deployment preparations.

**Prepared By:**

**Team Lead Approval:**

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**Uppara Ganesh**

Project Team Member

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**Boya Praveen**

Project Lead